

Interpreting Correlation and Least-Squares Regression

Reading the data, using the least-squares line, meaning of slope and intercept, and correlation

Directions:

- All problems use the study below. x is study time in hours; y is the test score in points.
- The least-squares line for these six students is $\hat{y} = 50 + 6x$.
- A residual is $y - \hat{y}$: actual minus predicted. Keep the sign.
- Round any decimal answer to four decimal places unless told otherwise.

Six students: study time and test score

Study time x (hours)	1	2	3	4	5	6
Test score y (points)	59	59	68	74	77	89

For this data set: $\sum(x - \bar{x})(y - \bar{y}) = 105$, $\sum(x - \bar{x})^2 = 17.5$, $\sum(y - \bar{y})^2 = 666$.

1. Read from the table: what score did the student who studied 4 hours earn?

Answer: _____

2. Use the least-squares line to predict the score of a student who studies 3.5 hours.

Answer: _____

3. The slope of the line is 6.

- (a) By how many points does the line predict the score changes for 2 additional hours of study? _____
- (b) Interpret the slope of 6 in the context of this study, in a sentence.

4. The student who studied 5 hours scored 77 points. Find the residual for that student.

Answer: _____

5. Using the three sums given above, compute the correlation coefficient $r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \cdot \sum(y - \bar{y})^2}}$.

Answer: _____

6. The coefficient of determination is r^2 .

- (a) Compute r^2 for this data set. _____
- (b) Interpret that number in the context of this study, in a sentence.

7. The intercept of the line is 50.

- (a) What score does the line predict for a student who studies 0 hours? _____
- (b) Explain whether that number is a trustworthy prediction for these data.

8. Consider the six students as a group.

- (a) One of the six is chosen at random. Find the probability that the chosen student studied more than 4 hours. Give an exact fraction. _____
- (b) The correlation here is about 0.97. Explain why that strong correlation does not show that studying causes the higher scores.

9. A teacher wants to use the line for a student planning to study 8 hours.

- (a) What score does the line predict at $x = 8$? _____
- (b) Explain why this particular prediction should be treated with caution.

10. Synthesis. A seventh student studies 4 hours and scores 62 points.

- (a) Find this student's residual against the existing line $\hat{y} = 50 + 6x$. _____
- (b) Explain what a residual of that size means, and how adding this point would change the correlation and the fitted line.